

Kosuke Ukita

PhD student (D1), full-time

Kyushu Institute of Technology

ukita.kosuke299@mail.kyutech.jp GitHub Scholar ORCID

EDUCATION

Apr. 2026 - Present	PhD in CSE Kyushu Institute of Technology Department of Creative Informatics, Graduate School of Computer Science and Systems Engineering
Apr. 2024 - Mar. 2026	MEng in CSE Kyushu Institute of Technology Department of Creative Informatics, Graduate School of Computer Science and Systems Engineering <u>Thesis: "A Study on Sensor Foundation Models via Joint Training using Flow Matching"</u> GPA: 3.6 / 4.0
Apr. 2020 - Mar. 2024	BEng in CSE Kyushu Institute of Technology Department of Artificial Intelligence, School of Computer Science and Systems Engineering <u>Thesis: "Proposal of Representation-Conditional Latent Diffusion Models: Hematoma Classification in Brain CT Images"</u> GPA: 3.7 / 4.0
Apr. 2017 - Mar. 2020	High School Diploma Kanonji Daiichi High School (Designated as a Super Science High School) Science and Mathematics Course GPA: 4.9 / 5.0

WORK EXPERIENCE

Apr. 2024 - Mar. 2027	Teaching Assistant (TA) Lecturer: Tsuyoshi Okita - Deep Learning Practicum (2026) - Advanced Topics in Deep Learning I (2026) - Deep Learning Practicum (2025) - Deep Learning (2024)
Dec. 2023 - Mar. 2026	Research Assistant (RA) Supervisor: Tsuyoshi Okita Conducted research and development on foundation models
Oct. 2023 - Nov. 2023	OPTiM Corporation Research Intern / Part-time Research Assistant Conducted research and development on AI technologies for DX in agriculture

SOCIAL EXPERIENCE

Nov. 2026	Student Volunteer (SV) The 14th International Symposium on Applied Engineering and Sciences (SAES 2026) · Fukuoka, Japan - Oral Session Chairperson - Other support
Jun. 2026	Student Volunteer (SV) The 30th Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD 2026) · Hong Kong, China - Registration desk - Session support - Excursion support - Other support

PUBLICATIONS

- [1]

Weight Tying Causes Grokking in Conditional Flow Matching

Kosuke Ukita

Tsuyoshi Okita

The 14th International Symposium on Applied Engineering and Sciences (SAES 2026), No. p0092

November 2026 · Fukuoka, Japan

Non-refereed

Oral

[2]

拡散モデルを用いた人間行動モーションデータからの写実的な映像生成

松石康希

浮田嵩祐

大北剛

情報処理学会論文誌 (IPSJ Journal), 67-10, 2026.

July 2026 · Japan

Refereed

Journal

[3]

Efficient Inference for Flow Matching via Unified Path CFG

Kosuke Ukita

Tsuyoshi Okita

The 30th Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD 2026)

June 2026 · Hong Kong, China

Refereed

Oral

[4]

High-Performance Self-Supervised Learning by Joint Training of Flow Matching

Kosuke Ukita

Tsuyoshi Okita

The 29th International Conference on Artificial Intelligence and Statistics (AISTATS 2026)

May 2026 · Tangier, Morocco

Refereed

Spotlight (top 3%)

[5]

From Sensor Data to Sequence of Scenes: Diffusion Model-Based Visualization of Human Activity

Koki Matsuishi

Kosuke Ukita

Tsuyoshi Okita

IEEE Xplore, International Conference on Activity and Behavior Computing (ABC 2026), pp. 1-10, DOI: 10.1109/ABC68169.2026.11567210

March 2026 · Hokaido, Japan

Refereed

Oral

[6]

Foundation Models to Tackle Activity Recognition in Unknown Domain: Sussex-Huawei Locomotion Challenge 2025 Task 2

Tsuyoshi Okita

Kosuke Ukita

Asahi Miyazaki

Daichi Kubota

Jukichi Ota

Naoki Kagiya

Asahi Nishikawa

Daichi Nagayasu

Syunya Tomitaka

Daisuke Nozaki

Yuki Odo

Raku Yamashita

Xiaolong Ye

Huayu Gao

Kazuki Okahashi

Koki Matsuishi

Masaharu Kagiya

Kodai Hirata

Haruki Kai

Lin Wang

Hristijan Gjoreski

Mathias Ciliberto

Paula Lago

Kazuya Murao

Daniel Roggen

Companion of the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp '25), pp. 983–991

October 2025 · Espoo, Finland

Refereed

Oral

[7]

Multimodal Foundation Model for Cross-Modal Retrieval and Activity Recognition Tasks

Koki Matsuishi

Kosuke Ukita

Tsuyoshi Okita

International Journal of Activity and Behavior Computing (IJABC), Care XDX Center, Kyushu Institute of Technology, Vol. 2025, No. 3, pp. 1-25

November 2025 · Kitakyushu, Japan

Refereed

Oral

[8]

Energy-Efficient High-Performance Representation Learning for Sensor Data via Flow Matching

Kosuke Ukita

Tsuyoshi Okita

The 7th International Symposium on Neuromorphic AI Hardware, Paper No. P-19

October 2025 · Kitakyushu, Japan

Non-refereed

Poster

[9]

Image Classification Using a Diffusion Model as a PreTraining Model

Kosuke Ukita, Xiaolong Ye, Tsuyoshi Okita

arXiv preprint arXiv:2505.06890, pp. 1-10

May 2025

Non-refereedPreprint

[10]

AURA-MFM: 詳細な行動理解のためのCLIPを用いたマルチモーダル基盤モデルの構築

松石康希, 浮田嵩祐, 大北剛

情報処理学会研究報告(IPSJ SIG Technical Report), Vol. 2025-UBI-85, No. 8

February 2025 · Osaka, Japan

Non-refereedOral

[11]

Improving Human Activity Recognition through Diffusion Models

Kosuke Ukita, Tsuyoshi Okita

The 12nd International Symposium on Applied Engineering and Sciences (SAES 2024), No. p0052

November 2024 · Fukuoka, Japan

Non-refereedOral

[12]

Ensemble Learning of Models for Human Activity Recognition

Asahi Miyazaki, Kosuke Ukita, Tsuyoshi Okita

The 12nd International Symposium on Applied Engineering and Sciences (SAES 2024), No. p0053

November 2024 · Fukuoka, Japan

Non-refereedOral

[13]

Analysis of Human Activity Recognition by Diffusion Models

Kosuke Ukita, Tsuyoshi Okita

Companion of the 2024 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp '24), pp. 458–463

October 2024 · Melbourne, Australia

RefereedOral

[14]

表現条件付き潜在拡散モデルと表現学習

浮田嵩祐, Ye Xiaolong, 大北剛

情報処理学会研究報告, 火の国情報シンポジウム 2024, No. A4-4, pp. 1-8

March 2024 · Kagoshima, Japan

Non-refereedOral

[15]

Towards LLMs for Sensor Data: Multi-Task Self-Supervised Learning

Tsuyoshi Okita, Kosuke Ukita, Koki Matsuishi, Masaharu Kagiya, Kodai Hirata, Asahi Miyazaki

Adjunct Proceedings of the 2023 ACM International Joint Conference on Pervasive and Ubiquitous Computing & the 2023 ACM International Symposium on Wearable Computing (UbiComp/ISWC '23 Adjunct), pp. 499–504

October 2023 · Cancun, Mexico

RefereedOral

GRANTS

- SPRING Scholarship Awardee

Japan Science and Technology Agency (JST)

April 2026 - March 2029 · JPY ¥1,800,000 (+ ¥900,000) / year
- Student Travel Award

The 30th Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD 2026)

June 2026 · USD \$500
- International Travel Grant

The Telecommunications Advancement Foundation

April 2026 · JPY ¥290,000

AWARDS

– Student Travel Award

The 30th Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD 2026)
June 2026

– Spotlight Presentation (Top 3%)

The 29th International Conference on Artificial Intelligence and Statistics (AISTATS 2026)
May 2026

– Promising Researcher Award

Information Processing Society of Japan (IPSJ) Kyushu Branch
March 2024

– Best Research Award

The Japan Statistical Society (JSS), Special Committee of Statistical Education (CSE),
The 8th Sports Data Analysis Competition
May 2019

SKILLS

LANGUAGES

Python · C++ · C · Java · JavaScript · TypeScript · Go · SQL · R

FRAMEWORKS & TOOLS

PyTorch · TensorFlow · Hugging Face · Jupyter · ABCI · Docker · Git

OTHER SKILLS

Deep Learning · Neural Networks · Model Training · Data Processing · API Development · Linux